

Activity No. <4.1>

<Arrays>

Course Code: CPE007

Program: Computer Engineering

Course Title: Programming logic and design

Date Performed: 9/9/2025

Section: CPE11S1

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6. Output

```

1222.cpp
1  #include <iostream>
2  using namespace std;
3
4  int main(){
5      int calcScores[10] = {90, 85, 78, 88, 92, 80, 75, 80, 89, 91};
6      cout<<"the value of the 5th element is:" <<calcScores[4]<<endl;
7
8      calcScores[0] =95;
9      cout<<"new value of the calcScores[0] is:" <<calcScores[0]<<endl;
10
11     for (int i = 0; i < 10; i++) {
12         cout<<calcScores[i]<<" ";
13     }
14
15     int var1 = 10;
16     int var2 = 20;
17     int temp = var1;
18     var1 = var2;
19     var2 = temp;
20     cout<<" " <<endl;
21     cout<<"var1:" <<var1<<endl;
22     cout<<"var2:" <<var2<<endl;
23
24     temp = calcScores[0];
25     calcScores[0] = calcScores[9];
26     calcScores [9] = temp;
27     for (int i = 0; i < 10; i++) {
28         cout<<calcScores[i]<<" ";
29     }
30 }
31 //bubble sort
32 int temp1, n=10, j;
33 for (int i = 0; i < n - 1; i++) {
34     for (int j = 0; j < j - 1; j++) {
35         if (calcScores[j] > calcScores [j + 1]) {
36             //swap scores
37             temp1 = calcScores [j];
38             calcScores [j] = calcScores[j + 1];
39             calcScores [j + 1] = temp1;
40         }
41     }
42 }

```

Compile Log
 Debug
 Find Results
 Close

```

-----
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\TIPQC\Documents\1222.exe
- Output Size: 1.76801776885986 MiB
- Compilation Time: 0.38s
    
```

1222.cpp

```
16  int var2 = 20;
17  int temp = var1;
18  var1 = var2;
19  var2 = temp;
20  cout<< " "<<endl;
21  cout<<"var1:" <<var1<<endl;
22  cout<<"var2:" <<var2<<endl;
23
24  temp = calcScores[0];
25  calcScores[0] = calcScores[9];
26  calcScores [9] = temp;
27  for (int i = 0; i < 10; i++) {
28      cout<<calcScores[i]<<" ";
29  }
30
31  //bubble sort
32  int temp1, n=10, j;
33  for (int i = 0; i < n - 1; i++) {
34      for (int j = 0; j < j - 1; j++) {
35          if (calcScores[j] > calcScores [j + 1]) {
36              //swap scores
37              temp1 = calcScores [j];
38              calcScores [j] = calcScores[j + 1];
39              calcScores [j + 1] = temp1;
40          }
41      }
42  }
43
44
45  //print sorted array
46  cout <<"scores in ascending order:" ;
47  for (int i = 0; i < n; i++) {
48      cout<< calcScores[i] <<" ";
49  }
50
51  cout<<endl;
52  return 0;
53
54 }
```

Compile Log Debug Find Results Close

```
-----
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\TIPQC\Documents\1222.exe
- Output Size: 1.76801776885986 MiB
- Compilation Time: 0.38s
```

```
C:\Users\TIPQC\Documents\1. X + v
the value of the 5th element is:92
new value of the calcScores[0] is:95
95 85 78 88 92 80 75 80 89 91
var1:20
var2:10
91 85 78 88 92 80 75 80 89 95
-----
Process exited after 0.01505 seconds with return value 0
Press any key to continue . . . |
```

```
int temp = var1;
var1 = var2;
var2 = temp;
cout<< " "<<endl;
cout<<"var1:" <<var1<<endl;
cout<<"var2:" <<var2<<endl;

temp = calcScores[0];
calcScores[0] = calcScores[9];
calcScores [9] = temp;
```

7. Supplementary Activity

8. Conclusion

I learned how to basic swapping and the use of temp.because before this lesson I didn't know how to use temp and the swap var but now after this lesson I kinda know how to use it properly and how it works. And I wanna make sure that this technique would help me to grow for the future.

9. Assessment Rubric